Applied Tech Program of Study — CVHS + CHS — 2027 and Beyond

Owners: Kirsch (CVHS, Engineering) · McAteer (CVHS, Industrial Design) · Macdonald (CHS, Computer Science) · Reinwald (Architecture) · Keim (CHS, Metals + Autos) **District CTE lead:** Nikki McFarland **Last review:** April 2026 **Status:** Working draft — reflects the committee-landed course list the live site exposes at `promptaisolutions.com/applied-tech/`.

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1. Purpose of this document

This is the internal planning document for the unified CVHS + CHS Applied Tech Program of Study, effective 2027. It pairs with the public-facing site as the canonical reference for:

- the course list and pathway structure
- the proficiency framework and industry-certification map
- the credit architecture (CTE default plus two dual-credit courses)
- the POS-completer requirement and how Work-Based Learning fits
- open gaps we're naming honestly (articulation, WBL coverage, teacher prep load)

The site answers "what would a student take?" The plan answers "how does this hold together as a Program of Study for Perkins V reporting?"

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2. Program shape

Three courses lock the 9th-grade entry layer. Every Applied Tech student takes at least one, at either building:

- Computer Skills (Randy Macdonald)
- Woods 1 (Adam Kirsch, Andy McAteer, Stephanie Reinwald)
- Metals 1 (Andy McAteer, Stephanie Reinwald, JJ Keim)

Nine pathways span 10th–12th grade. Four are year-by-year sequences where each course

builds on the previous; five are collections of peer courses that can be taken in any order.

LADDERS (4)

Pathway	Site	Courses	Teacher
Computer Science	CHS	Intro to CS → Advanced CS → Capstone CS	Macdonald
Automotive	CHS	Autos 1 → Autos 2 → Autos 3	Keim
Metals	CHS	Metals II → Metals III	Keim
Industrial Design	CVHS	Industrial Design I → Industrial Design II	McAteer (+ Macdonald on I)

COLLECTIONS (5)

Pathway	Site	Courses	Teachers
Engineering	CVHS	Intro to Engineering · Engineering Science	Kirsch · Reinwald · McAteer · Keim
Woods & Carpentry	CVHS	Woods II · Carpentry	Kirsch
Architecture	CVHS	Architecture	Kirsch · Reinwald
Vehicular / Drone Design	CHS	Vehicular / Drone Design	Keim
Real World Math	CVHS	Real World Math	Keim · Kirsch

Total: 3 freshman courses + 17 upper-level courses = **20 courses** across 5 teachers and 2 buildings. The live site renders this map.

Metals III, added April 2026. The committee added Metals III as the advanced welding year at CHS (AWS SENSE Level 1 + TIG) to close the Metals pathway to a three-credit POS completer — previously, a Metals-focused student had no in-pathway route to completer without cross-counting. Metals III is JJ Keim's course and flips the Metals pathway from a single-course collection to a two-course ladder.

WHAT THIS REPLACES

Earlier drafts of this POS proposed IDEA → STEAM → DREAM as the CVHS design lane, a new "Shop Foundations" intro, and kept Small Engines as a standalone. The committee elected a simpler structure that names freshman shop courses directly (Woods 1, Metals 1) rather than collapsing them into a blended intro, and folded Small Engines into the Autos 1–3 sequence. The current model is what is reflected on the site; this document tracks only that model.

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3. Credit architecture

Every year-long Applied Tech course carries **1.0 CTE credit**. Two courses also carry a secondary credit, awarded on demonstrated proficiencies rather than by seat-time:

- **Engineering Science** — 0.5 CTE + 0.5 science. Physics-of-design anchored in measurement, forces, materials, and a documented design/test cycle.
- **Real World Math** — 0.5 CTE + 0.5 math. Shop-and-industry mathematics: measurement, layout trig, estimating, spreadsheet modeling.

Every other course is CTE-primary only. There is no automatic double-dipping — if a proposed new course wants a secondary credit tag, it must name the proficiencies that earn it.

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4. Proficiency framework

Framework: Hybrid — Oregon CTE TSA spine, with one embedded industry certification per intermediate or advanced course serving as the "meets" threshold.

This framework satisfies Perkins V TSA requirements (state-native rubric) and gives students a portable artifact (the industry cert) they can take to LBCC or an employer. Three-course ladder pathways stack their certs; collection pathways hold a cert per course.

CANDIDATE CERTIFICATION MAP

Course	Candidate certification
Computer Skills	Typing benchmark + digital-citizenship rubric (no external cert)

Course	Candidate certification
Woods 1	OSHA-10 General Industry + shop-safety TSA
Metals 1	OSHA-10 General Industry + shop-safety TSA
Intro to CS	ODE CS Skill Set (no commercial cert at this level)
Advanced CS	AP CS Principles eligibility OR Codecademy portfolio
Capstone CS	Portfolio review + WBL artifact
Intro to Engineering	CSWA (SolidWorks Associate)
Engineering Science	Oregon TSA Engineering assessment
Architecture	Autodesk Certified User — Revit
Woods II	NCCER Carpentry Level 1 (wood-focused modules)
Carpentry	NCCER Carpentry Level 1 + documented jobsite hours
Metals II	AWS SENSE Entry Welder; GD&T module from NIMS
Metals III	AWS SENSE Level 1 (SMAW / GMAW / FCAW) + TIG (GTAW) competency
Autos 1	ASE Entry-Level Student Certification
Autos 2	EPA 609 (MVAC refrigerant handling)
Autos 3	ASE Entry-Level Student Certification (second pathway)
Industrial Design I	CSWA (SolidWorks Associate)
Industrial Design II	CSWP (SolidWorks Professional) or portfolio review
Vehicular / Drone Design	FAA Part 107 prep (for drone track) or build-log artifact (for vehicle track)

Course	Candidate certification
Real World Math	None — dual-credit math-rubric already anchors it

Every course also carries 4–6 measurable proficiency statements in the site's data model (`js/data.js`, `proficiencies` array). The committee has drafted proficiencies for every course; a polish pass is outstanding and the relevant courses carry a **Draft** chip on the site until that pass lands.

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5. POS-completer requirement

A completer demonstrates, per Oregon Perkins V rules:

1. At least **3.0 credits** in the pathway.
2. At least **0.5 credit** at the intermediate or advanced level (10th grade and above in our implementation).
3. A recorded **Work-Based Learning** experience.
4. The **Technical Skill Assessment** (Oregon CTE TSA, scored to an industry standard).

The site's per-course grade level and credit metadata are what a counselor reads to verify (1) and (2). (3) and (4) require the supporting infrastructure in §6 and §4.

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6. Work-Based Learning status

A known gap we are naming honestly. The site's district page has a full table; the short version:

- **In place:** Carpentry (Unity Shelter tiny-home builds generate documentable jobsite hours).
- **Planned:** Automotive (Autos 3 capstone repair/restoration), Vehicular / Drone (capstone build + run log), Computer Science (Capstone CS showcase — needs a community/employer partner to qualify), Industrial Design II (self-directed project — needs a client partner).
- **Planned → Metals:** with Metals III added (April 2026), the AWS SENSE Level 1 capstone fabrication is designed to generate a documentable project with industry-standard weld map + bill of materials. Pairing this with a local metalwork shop for a supervised fabrication would close Metals to full WBL.
- **Gap:** Engineering, Architecture — no WBL embedded today. A partnership with an

engineering firm or a local architecture practice is the obvious closure.
- **Not applicable:** Real World Math (credit-bearing support course).

WBL closure is a resource ask on the district, not a curriculum decision. Teacher time and industry relationships are the constraint.

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7. LBCC College Now articulation

The thinnest part of the program and we are not pretending otherwise.

Course	Candidate LBCC equivalent	Status
Intro to Engineering	EG 4.407 (historical IDEA articulation)	Needs re-articulation for the renamed course
Computer Science series	CS 160 / CS 161	Not yet articulated
Industrial Design I / II	MFG / CADD programs	Not yet articulated
Autos 1 / 2 / 3	Automotive Technology program	Not yet articulated
Metals II	Welding or Machine Tool programs (intro level)	Not yet articulated
Metals III	WLD 121 / 122 candidates (AWS SENSE Level 1 alignment)	Not yet articulated
Woods II / Carpentry	Construction Tech / NCCER	Not yet articulated
Engineering Science, Architecture, Vehicular / Drone, Real World Math	—	No current crosswalk identified

Re-articulation is a committee-and-LBCC conversation to schedule once the 2027–28 catalog lands. Until then, advising conversations should treat each course as "candidate," not "articulated."

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8. Teacher-prep sustainability

Raw counts, no judgment, per committee decision:

Teacher	Courses	Distinct pathways
Adam Kirsch	7	4
JJ Keim	9	5
Andy McAteer	5	3
Randy Macdonald	4	2
Stephanie Reinwald	4	2

With Metals III added (April 2026), JJ Keim now carries 9 courses — the highest course load of any teacher. His pathway count held at 5 (Metals III is within the existing Metals pathway). The committee should watch this closely in year one.

The ≤3-preps-per-teacher target the earlier draft of this plan proposed is **not achievable** under the current staffing and course list. Kirsch and Keim especially are carrying four and five pathways respectively. The committee has deferred the sustainability rule until the first year of implementation reveals which prep combinations are actually onerous versus merely distinct. The district should expect a prep-relief ask in year one.

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9. Decisions landed

The committee has signed off on:

- The 19-course model above (freshman row + 9 pathways).
- The CVHS / CHS site split (CVHS = Engineering, Architecture, ID, Woods + Carpentry, Real World Math; CHS = CS, Autos, Advanced Metals, Vehicular/Drone).
- Dual-credit assignments (Engineering Science for science, Real World Math for math).
- Pathway groupings and their type (ladder vs. collection).
- Which teacher leads each course.

10. Open questions

Still under discussion, in rough priority order:

1. **Prerequisite chains above the freshman row.** The committee has chosen to ship with *no* hard prereqs above 9th grade — all upper-level sequencing is currently "suggested." This may tighten for specific ladder steps (e.g., Intro to CS → Advanced CS) once the teaching team has a year of data.
2. **Industry certification funding.** Perkins Reserve funds are the likely source; the district needs to approve cert-exam seats as a line item.
3. **WBL closure for Metals / Engineering / Architecture** (see §6).
4. **LBCC re-articulation** (see §7). Nothing ships without LBCC on the other side of the conversation.
5. **Prep-load sustainability rule** (see §8). Blocked pending year-one data.
6. **Cross-site scheduling.** Committee consensus that a student may take one course per year at the other building. Transportation and bell-time alignment are real constraints the district needs to address.

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11. Candor pass

Naming the fragile parts on purpose:

- **Teacher prep load is above the historical comfort line.** Kirsch's 4 pathways and Keim's 5 are not sustainable long-term; the plan runs on goodwill in year one.
- **WBL is only credibly in place for Carpentry.** Every other pathway's WBL story is aspirational, not documented.
- **LBCC articulation is essentially a blank slate.** Historic IDEA credit doesn't automatically transfer to the renamed Intro to Engineering — we will have to earn it again.
- **The "no hard prereqs above 9th grade" rule** makes advising easier and reduces student lock-in, but it also means Advanced CS students may arrive without Intro to CS background and Capstone CS may see unevenly-prepared students. Teachers will absorb that variance in year one.
- **Small Engines was folded into the Autos sequence.** Students whose interest is specifically powersports or ag-mechanics no longer have a named course — they get what Autos 1–3 provide, which is broader but less specialized.
- **The whiteboard PST / IDEA / STEAM / DREAM vocabulary is gone** from the student-facing site. Faculty who knew those courses by those names will need to map onto

the new names. The site's About page explains the rename; the committee should reinforce it at the first 2026–27 department meeting.

Appendix — reference files

- **Live site:** https://promptaisolutions.com/applied-tech/
- **Source:** https://github.com/mr-mcateer/applied-tech — all course/teacher/pathway content lives in `js/data.js`
- **Working documents:** `v2-work/design-spec.md`, `v2-work/content-plan.md`, `v2-work/ref-candidates.md` (the godly.website reference scout)
- **Earlier artifacts (retained for context):**
- `Applied Tech Pathways 2027+ — for Nikki McFarland.pptx` (committee deck, pre-site-build)
- `~/Downloads/Applied Tech Pathways — Flow Chart.pdf` (one-page student-facing flowchart)
- `CVHS-Course-Handbook-24-25.pdf` (source for Metals 1, Metals 2, Carpentry, Architecture descriptions)
- `Engineering Pathway and Course Descriptions - FILLED.docx` (source for the engineering-lane heritage courses)

Edits to the canonical content go into `js/data.js` via the public repo. Edits to this plan go to whichever member of the committee has the active draft.